

Akshat Kalra

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EDUCATION

The University of British Columbia

Vancouver, BC

BSc Computer Science and Statistics | *Dean's Honor List*

Expected Graduation May 2027

- **Relevant Coursework:** Computer Networking (97%), Statistical Modelling (95%), Software Construction (91%), Models of Computation (93%).
- **Scholarships:** Faculty of Science International Student Scholarship (2025, 2024), OIS Scholarship (2022).
- **Leadership / Activities:** Notion Campus Leader, VP Internal - Brock Commons Residence Association.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Rust, TypeScript, **SQL (MySQL, PostgreSQL)**, HTML/CSS, R, Racket (Lisp Dialect)

Frameworks: React.js, Next.js, Nest.js, Node.js, Flask, JUnit, FastAPI

Developer Tools: Git, Docker, AWS, VS Code, **Linux Operating System**

Networking: Familiar with networking concepts such as TCP/IP, QUIC Protocol, OSPF, BGP, DHCP, IP VPN, VLAN.

EXPERIENCE

Software Developer (Co-op)

May 2025 – Present

Faculty of Forestry, University of British Columbia

Vancouver, BC

- Engineered CanvasLMS-to-pgvector data pipeline integrating discussions, pages, files, modules and quizzes for a web application used by 1500+ students and multiple courses at UBC. [Parent Issue Link](#)
- Built resource selector enabling professors to control LLM access by resource type, quiz questions, and metadata filters. [Issue Link](#) & [PR Link](#)
- Designed **TypeORM-PostgreSQL** models and NestJS **REST API** endpoints for Canvas integration, managing embeddings lifecycle and data synchronization.
- Architecting AWS Bedrock RAG system with Python/LangChain to automate grading with citation-backed AI responses.
- **Code contributions:** [Link to all my PRs & assigned issues](#)

Undergraduate Teaching Assistant

Sep 2023 – Present

Department of Computer Science & Philosophy, UBC

Vancouver, BC

- 4x Teaching Assistant for **CPSC 121** (Models of Computation): led weekly discussions/labs for 100+ students
- 2x Teaching Assistant for **PHIL 220** (Symbolic Logic): graded 200+ assignments & led weekly office hours.

PROJECTS

TCP Implementation | C, Linux Socket APIs, Network Protocols

Mar 2025

- Developed a server with connection setup/teardown, congestion control, and fast retransmission..
- Implemented sender-side flow control, retransmission timers, and reliable file transfer with checksum validation under packet loss.

Tranquilo | Next.js, Flask, SQLite, BERTopic, NLP

Jan 2025

- Built RAG-based mental health journal using 3,500+ row therapist-response dataset
- Applied BERTopic (NLP) for topic modeling and stored embeddings in a Vector DB for fast retrieval and pattern discovery.
- Architected microservices & data pipelines, optimizing orchestration to reduce response time from 15+ seconds to 2–4 seconds (73% improvement) through performance profiling and bottleneck analysis.

Airbnb Policy Prediction | *1st Place at Case Competition* | Scikit-learn

Nov 2024

- Built end-to-end ML pipeline using Random Forest regressor that **won \$300 and first place** at the competition.
- Conducted feature engineering, hyperparameter tuning, and model evaluation, reducing MSE from 1.64M to 1.19M.
- Conducted deep-dive **exploratory data analysis** to uncover hidden patterns in housing policy impacts; documented methodology and insights in [Medium article](#), demonstrating data storytelling to non-technical audiences.

TECHNICAL EXTRACURRICULARS

Hackathons & Case Competitions

- UBC WiDS Case Competition (**Winner**), UBC GDSC × Launchpad Hackathon (**Winner**), UBC CIC × AWS GenAI & Sustainability Hackathon (**3rd Place**), nwPlus cmd-f 2025 (**Mentor & Judge**), nwPlus HackCamp 2025 & 2024 (**Mentor**), ASA Datafest 2025 (**Mentor**)

Autonomous Underwater Vehicle – UBC Subbots | C++, ROS2, Python

Sep 2024 – Jan 2025

- Part of the engineering team designing an AUV competing in the RoboSub.
- Worked on a containerized dev environment (**Linux Ubuntu 20.04**, ROS2) using **Docker**, intended to reduce onboarding time for new members and improving development workflow efficiency.